

# Measuring cubic norm-one class admission over 192–252-bit fields

ISO-1 research evidence

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## Abstract

We independently sample 512 full-2-torsion curve models in ten finite fields of bit lengths 192–252 and extension degrees 6, 10, and 14. The necessary conductor condition admits 323 of the 512 ordinary curves. Every admitted curve has a verified full-4-torsion representative at distance at most one rational 2-isogeny, after choosing the trace sign. We prove that this torsion property is exactly what the admission condition characterizes. The bounded degree-2/3 searches find 0 norm-one weak witnesses among these independent starts; 323 admitted class labels remain unresolved. All ten separately constructed positive controls recover their known weak class. A norm-fiber count also bounds the probability that a uniform source model is already weak by  $3/(p^2 - 1)$ , separating direct model density from class reach. The class labels concern the cubic norm-one branch. A separate exact quartic control demonstrates that the quadratic branch in Joux–Vitse can occupy a class empty in the cubic census.

## 1 The measured population

Put  $q = p^2$ ,  $K = \mathbf{F}_{q^n}$ ,  $Q = q^n$ , and  $N = (Q - 1)/(q - 1)$ , with odd  $n \geq 3$ . The weak family is  $y^2 = x(x - \alpha)(x - \alpha^q)$ , up to  $K$ -isomorphism. An ordinary weak class must satisfy

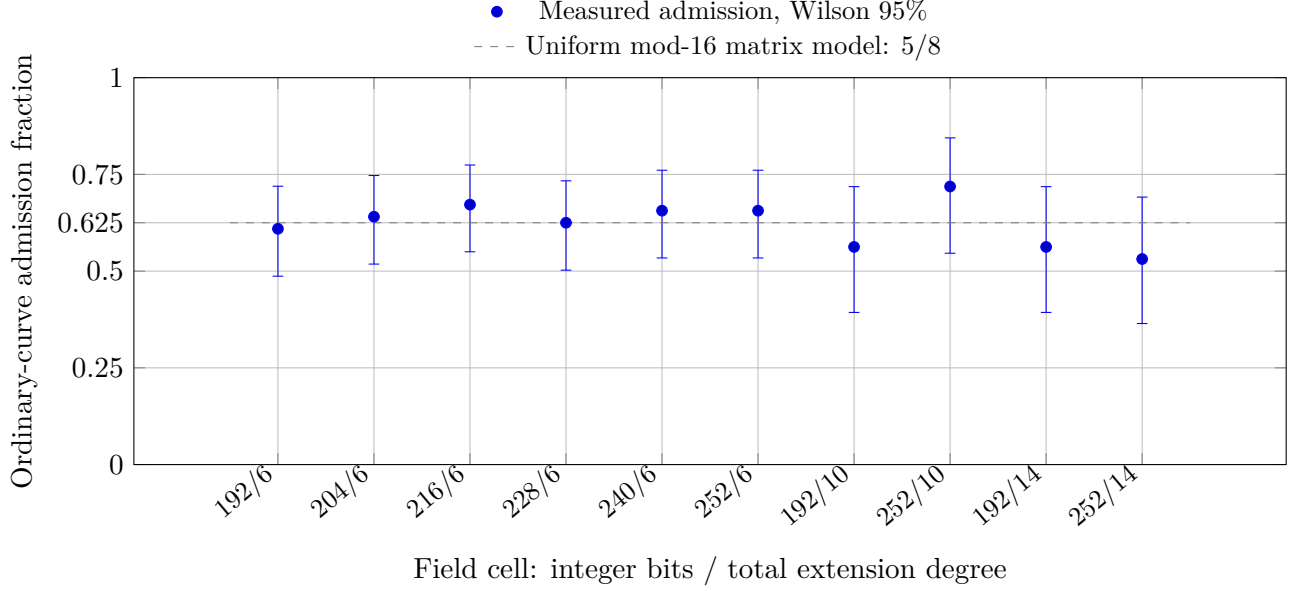
$$t \equiv \pm(Q + 1) \pmod{16}, \quad 4 \mid f_\pi,$$

where  $f_\pi$  is the conductor of the Frobenius order, separately from the endomorphism-order conductor. The previous theorem proves this condition in every such odd extension. The new measurement samples  $(u, v)$  uniformly among distinct nonzero elements of  $K$  and counts  $E : y^2 = x(x - u)(x - v)$ . Consequently the population is uniform in these ordered root pairs and includes both quadratic twist signs; it is not uniform in traces or in isogeny classes. Equal traces remain separate rows.

| Bits | Degree | $p$           | Samples | Admitted | Rate (Wilson 95%) |
|------|--------|---------------|---------|----------|-------------------|
| 192  | 6      | 4294967291    | 64      | 39       | 60.9% (48.7–71.9) |
| 204  | 6      | 17179869143   | 64      | 41       | 64.1% (51.8–74.7) |
| 216  | 6      | 68719476731   | 64      | 43       | 67.2% (55.0–77.4) |
| 228  | 6      | 274877906899  | 64      | 40       | 62.5% (50.3–73.3) |
| 240  | 6      | 1099511627689 | 64      | 42       | 65.6% (53.4–76.1) |
| 252  | 6      | 4398046511093 | 64      | 42       | 65.6% (53.4–76.1) |
| 192  | 10     | 602233        | 32      | 18       | 56.2% (39.3–71.8) |
| 252  | 10     | 38543917      | 32      | 23       | 71.9% (54.6–84.4) |
| 192  | 14     | 13421         | 32      | 18       | 56.2% (39.3–71.8) |
| 252  | 14     | 262139        | 32      | 17       | 53.1% (36.4–69.1) |

Intervals are Wilson 95% intervals for independently sampled curve admission. Bit length means  $\lfloor \log_2 Q \rfloor + 1$ ; all fields meet the requested integer bit sizes. Six degree-6 primes are the largest primes

below  $2^{32}, 2^{34}, \dots, 2^{42}$ . The degree-10 and degree-14 endpoint primes are the largest below the corresponding integer root of  $2^b - 1$ . Exact cardinalities, irreducible moduli, coefficients, full ICV1 identities, and raw hashes are in [the records](#).



The dashed reference is an exact count under a specified matrix model: among the 512 matrices modulo 16 congruent to the identity modulo 2 with determinant  $Q \bmod 16 \in \{1, 9\}$ , exactly 320 have the admitted trace. It is not a proved finite-field population law. Each field is its own sampling stratum; the pooled admission is a summary of this fixed design.

## 2 What the conductor condition characterizes exactly

**Theorem 1** (A full-4-torsion representative within one edge). *Let  $K$  have odd cardinality  $Q \equiv 1 \pmod{4}$ , and let  $E/K$  have full rational 2-torsion. Then  $16 \mid \#E(K)$  if and only if  $E$  itself, or one of its rational degree-2 quotients, has full rational 4-torsion. Thus  $t \equiv \pm(Q+1) \pmod{16}$  is equivalent to this property for one of the two quadratic twist signs.*

*Proof.* A full-4-torsion representative has group order divisible by 16, and an isogeny preserves the order. Conversely, the 2-primary group of an elliptic curve has at most two cyclic factors. If  $E[4](K)$  is not full and  $16 \mid \#E(K)$ , full rational 2-torsion forces a rational point  $P$  of order 8. Put  $T = 2P$  and  $A = 2T = 4P$ . Translate  $A$  to  $(0, 0)$  and write

$$E : y^2 = x(x-u)(x-v) = x(x^2 + ax + b), \quad a = -(u+v), \quad b = uv.$$

The duplication formula is

$$x(2R) = \frac{(x(R)^2 - b)^2}{4y(R)^2}.$$

Since  $2T = A$ , write  $x(T) = s$  with  $s^2 = b$ . Applying the translated duplication formula at each rational root shows that  $x(2P) - e_i$  is a square for all three roots  $e_i$ . In particular,  $s = x(T)$  is a nonzero square. Since  $T$  lies on  $E$ , also  $a + 2s = y(T)^2/s^2$  is a nonzero square.

The degree-2 quotient by  $A$  has equation

$$E' : y^2 = x[x^2 - 2ax + (a^2 - 4b)]$$

and roots  $0, a + 2s, a - 2s$ . The product of the two nonzero roots is  $(u - v)^2$ , so both roots are squares. Their difference is  $4s$ , also a square. Because  $-1$  is a square in  $K$ , every signed root difference is a square. The standard rational halving formulas give independent halves of two distinct 2-torsion points, so  $E'[4]$  is full. For the trace pair, one order is divisible by 16 exactly under the displayed congruence, and both twists retain full rational 2-torsion.  $\square$

The cleared duplication identity has record `IDC1h40099ec00a503c89`; the earlier rational-halving and degree-2-map identities retain their existing certificates. An independent exhaustive check covers all 5,264 ordered root pairs in eight fields of cardinalities 5 through 49. It independently factors each quotient's 2-torsion polynomial and point-counts each quotient. All 1,336 order-divisible-by-16 cases agree with the one-edge equivalence; 592 already have full 4-torsion and 744 require an edge. The exact output is retained in `torsion_exact.txt`. Every admitted large-field sample passes explicit halving and map checks. This theorem explains the exact torsion feature observed at large field sizes. It leaves the additional norm-one class-intersection condition as a distinct question.

### 3 Direct weak-model density

**Theorem 2** (Norm-fiber bound). *For the sampled full-2-torsion model, put  $\lambda = v/u$ . Then  $\lambda$  is uniform in  $K \setminus \{0, 1\}$ , and*

$$\Pr(E \text{ already has a weak model}) \leq \frac{3(N-1)}{Q-2} < \frac{3}{q-1}.$$

*Proof.* Writing  $\mathcal{N} = N_{K/\mathbf{F}_q}$ , the three choices of translated root give the exact direct-model conditions

$$\mathcal{N}(\lambda) = 1, \quad \mathcal{N}(\lambda - 1) = -1, \quad \mathcal{N}((\lambda - 1)/\lambda) = 1.$$

For example, equality of the norms of  $u$  and  $v$  is the first condition. Hilbert 90 gives the required conjugate pair; multiplying its parameter by an element of  $\mathbf{F}_q^\times$  selects the square class needed for a  $K$ -isomorphism, since the extension degree is odd. The other two choices give the remaining conditions. Each event has  $N - 1$  parameters: the norm map has fibers of size  $N$ , and the corresponding linear or fractional-linear substitution removes exactly one excluded parameter from that fiber. The union bound proves the first inequality. Since  $(q - 1)N = Q - 1$ , its numerator gives  $3(N - 1)/(Q - 2) = 3(Q - q)/[(q - 1)(Q - 2)] < 3/(q - 1)$ .  $\square$

The geometric-sum identities at odd degrees 3, 5, and 7 have records `IDC1hc1132e34bb8c94e8`, `IDC1h547a8ae22b837173`, and `IDC1hcacc5ce8020b68ac`. Each certificate passes 32 recorded plus 64 additional evaluations; exact GP symbolic expansion returns zero. Exhaustive direct-locus checks cover 15,623, 117,647, and 59,047 parameters at  $(p, n) = (5, 3), (7, 3), (3, 5)$ , respectively. Each of the three event counts is exactly  $N - 1$ .

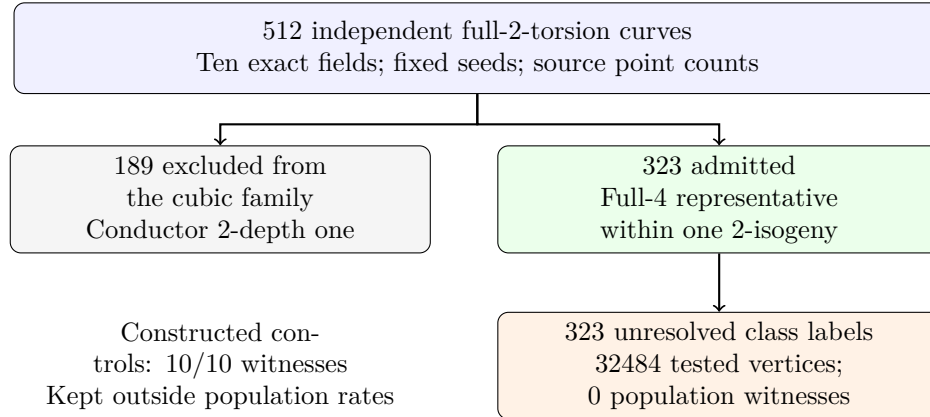
| Bits | Degree | $p$           | Direct-model probability upper bound |
|------|--------|---------------|--------------------------------------|
| 192  | 6      | 4294967291    | $1.626 \times 10^{-19}$              |
| 204  | 6      | 17179869143   | $1.016 \times 10^{-20}$              |
| 216  | 6      | 68719476731   | $6.353 \times 10^{-22}$              |
| 228  | 6      | 274877906899  | $3.970 \times 10^{-23}$              |
| 240  | 6      | 1099511627689 | $2.482 \times 10^{-24}$              |
| 252  | 6      | 4398046511093 | $1.551 \times 10^{-25}$              |
| 192  | 10     | 602233        | $8.272 \times 10^{-12}$              |
| 252  | 10     | 38543917      | $2.019 \times 10^{-15}$              |
| 192  | 14     | 13421         | $1.666 \times 10^{-8}$               |
| 252  | 14     | 262139        | $4.366 \times 10^{-11}$              |

This is a bound on the randomly drawn source model, separately from the presence of a weak representative anywhere in its isogeny class. Increasing the odd relative degree while fixing the total bit size makes the base field smaller and increases this direct-model density bound.

## 4 Witness searches and the remaining class labels

All witness and exclusion labels here select the displayed cubic norm-one family. The broader quadratic branch is tested separately below. For relative degrees 5 and 7, the rows verify generalized norm-one arithmetic and torsion; the cited genus-3 cover applies to relative degree 3. For each admitted ordinary sample, the search follows rational isogenies of degrees 2 and 3, retaining vertices with full rational 2-torsion. The fixed limits are 256 distinct  $j$ -invariants, 90 seconds for the search, and 240 seconds for the complete GP child. Each retained edge checks its 2-torsion roots and three mapped points. A witness must satisfy the exact norm equalities and pass a fresh point count with the selected source trace. The two trace signs share weak-class presence because the family is closed under quadratic twisting.

The population run tests 32484 vertices and evaluates 54216 retained degree-2 and 38093 degree-3 edges. It finds 0 witnesses. The 323 admitted classes remain unresolved: 219 searches exhaust their restricted component and 104 reach the vertex cap. There are 0 search-time caps and 0 failed or parent-capped cells. Exhausting this restricted graph does not enumerate the whole isogeny class. With no exact labels for these admitted large-field classes, the identified precision interval is  $[0, 1]$ . The previously observed small-prime census precision therefore remains a measurement of its own population.



All ten constructed controls start off the direct weak locus, at a 2-isogeny neighbor of a norm-one curve. The search finds a weak endpoint after testing two vertices in each case. Both source and endpoint are point-counted independently. These controls demonstrate witness detection at the requested sizes and are excluded from population proportions. The source and endpoint of one 252-bit control, including full ICV1 identities and the kernel, are shown in [the evidence-linked diagram](#).

The small-field validation has 64 independent  $\mathbf{F}_{76}$  source models. The complete exact census labels 42 of their classes weak and 22 zero. Of 49 admitted starts, the search finds 38 witnesses. Among 11 restricted component closures, seven are exact zeros and four are weak classes that this restricted search misses. All 15 rejected classes are exact zeros. This supplies an explicit validation of the unresolved-label policy. Four additional constructed small-field controls also succeed.

## 5 Reproducibility and requirement status

The pre-run [protocol](#) fixes the sample law, seeds, fields, caps, and label rules. A compiled Rust driver runs four PARI/GP 2.17.3 workers. The host is an Apple M4 Pro, 14 logical CPUs, 48 GiB memory, macOS Darwin 25.6.0 arm64. The source hash, native executable hash, GP executable hash, command

parameters, exit status, wall resource receipt, and both raw output hashes are retained for every attempted cell. Resource times are from a shared host and do not constitute controlled timing ratios. The native checker rebuilds summary tables and canonical model identities from the raw records and rejects changed receipts or labels. An additional GP replay reconstructs every model from the recorded modulus and roots, checks its short coefficients and  $j$ , and verifies all saved witness routes.

| Requested requirement                       | Current evidence status                                                                                   |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 192–252-bit fields and larger degrees       | Verified: 512 counted samples in ten fields, degrees 6, 10, 14.                                           |
| Necessary condition and torsion measurement | Verified: 323 admissions and matching full-4 representatives.                                             |
| Explicit weak-model search                  | Verified bounded result: population 0, constructed controls 10/10.                                        |
| Large-field weak-class prediction precision | Partial: 323 admitted labels unresolved; exact CM support construction remains the missing scalable step. |
| Theorem and reproducible report             | Proofs, four algebra records, native checks, raw receipts, editable visuals, and this PDF.                |

The existing cubic-family CM norm-one criterion gives a necessary-and-sufficient class decision by intersecting the norm-one locus with selected CM orders. Its order-support construction is separate from the constant-degree norm checks. A recorded preflight on the first admitted 192-bit source obtains  $f_\pi = 8$  and a 188-bit fundamental discriminant, then `polclass` stops with `overflow in t_INT-->long assignment`. This is an observed implementation limit in the exact support path; its raw output, error, and fail-closed receipt are retained in `cm_preflight.*`. The next algorithmic step is an on-demand CM intersection or a certified class-action traversal at these bit sizes, followed by exact positive and zero labels for independently sampled admitted classes. The earlier complete small-prime census continues under its finite local worker; it is not pooled into this experiment.

**Primary implementation reference.** PARI’s [elliptic-curve documentation](#) specifies finite-field `ellcard`, polynomial-kernel `ellisogeny`, and `ellisogenyapply`. The program retains each witness kernel and the actual target model produced by the polynomial map, so the route can be replayed without choosing another isomorphic model.

## 6 Contribution relative to Joux–Vitse

The cover-and-decomposition method is credited to [Joux–Vitse](#). Their Section 4.1 includes both linear and quadratic  $h \in \mathbf{F}_q[x]$  in  $y^2 = h(x)(x - \alpha)(x - \alpha^q)$  and discusses class reach heuristically. Our cubic norm-one theorem and census concern the linear branch.

We tested 100 nonsplit quadratic models over  $\mathbf{F}_{76}$  with fixed  $d \in \mathbf{F}_{49}$  nonsquare. Forty-six ordinary traces fail the cubic conductor predicate. Complete affine- $x$  enumeration, with two infinity points, independently verifies three selected models:

| Trace | Exact cardinality | $f_\pi$ | $v_2(f_\pi)$ |
|-------|-------------------|---------|--------------|
| −38   | 117,688           | 18      | 1            |
| −10   | 117,660           | 26      | 1            |
| 610   | 117,040           | 72      | 3            |

Thus the exact cubic zero at trace  $\pm 610$  has a published-family quadratic representative. It is not an empty class for the full family. The equations, parameters, exact counts, and full model identities are retained in [the three quadratic records](#).

The standard branch-point coordinate change sets  $c_3 = f'(\alpha)$ ,  $c_2 = f''(\alpha)/2$ ,  $c_1 = f'''(\alpha)/6$  and

$$X = \frac{c_3}{x - \alpha}, \quad Y = \frac{c_3 y}{(x - \alpha)^2}, \quad Y^2 = X^3 + c_2 X^2 + c_3 c_1 X + c_3^2.$$

Each control checks the inverse and three mapped points. The cleared coordinate identity has record `IDC1hb01002991ea88fb4`. Seventy-five retained cubic positive models also have constructive Hilbert–90 parameters and explicit coordinate changes verified during the independent replay.

The contribution to develop is the subfamily’s conductor restriction, exact class evidence, and orbit census formulas. The one-edge torsion explanation uses standard arithmetic: the root-difference halving criterion is already in [Auer–Top](#), Lemma 2.1 and Proposition 2.1. Priority for the norm-one application against the wider cover literature remains to be checked. The next stronger result is a complete two-branch criterion with an explicit construction and a verified cost on independently sampled classes. [The attribution and scope comparison](#) gives the evidence and remaining novelty checks.